Analytical Cume (2) March 31, 2017

1. (10 pts) Write the reaction showing the dissolution of ferrous sulfide.

2. (10 pts) Write the solubility product expression, the constant for the solubility product for FeS at 25 °C is 8.0×10^{-19}

3. (10pts) The standard reduction potential of the half-reaction shown below is –0.440 V.

$$Fe^{2+} + 2e^- \leftrightarrow Fe$$

   Apply the Nernst equation of this half reaction indicating how concentration effects the potential, or E.

4. (40 pts) By combining the thermodynamic expressions above. Calculate E° for the following half-reaction, (using base 10 log, @ 25 °C, $\frac{RT}{F}$ = 0.0591 V/mole e−)

   $FeS + 2e^- \leftrightarrow Fe + S^{2-}$

5. (10 pts) Comment on why there is a difference in E° when reducing Fe(II) out of the solid FeS, rather than Fe(II) in solution.

6. (10 pts) Construct the line notation of a cell that consists of a saturated calomel reference electrode at the left hand, and an iron indicator electrode coated with insoluble FeS, which can be used to measure the S^{2-} concentration of a solution at the right hand.

7. (10pts) If the cell described by the line notation above has a potential of -1.057 V, what is the pS of the solution? (E_{SCE} = 0.241 V)